

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Summer Examination – 2023

Course: B. Tech.

Branch : E&TC

Semester : VI

Subject Code & Name: BTETPE603A- Microprocessors and Microcontrollers

Max Marks: 60

Date: 17/07/2023

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q.1 Solve Any Two of the following.		12
A) Provide a detailed explanation of the flag register of an 8085 microprocessor.	L6	6
B) State addressing modes of following instructions of 8085 microprocessor. up: LXI H, 1000H MOV A, M MOV B, A ADD B STA 1001H JMP up	L3	6
C) State any six differences between CISC and RISC architecture.	L3	6
Q.2 Solve Any Two of the following.		12
A) Explain following instructions with suitable examples of each. (i) ANI 55H (ii) CMP B (iii) SBI 55H	L6	6
B) Assume that five numbers are stored from 1000H to 1004H. Write an assembly language program to clear D3 bit of each number.	L4	6
C) Draw timing diagram for following instructions of 8085. (i) MOV A, B (ii) MVI A, 55H	L5	6

Q. 3 Solve Any Two of the following.		12
A) Draw and explain Pin diagram of an 8051 microcontroller.	L6	6
B) Explain any three addressing modes of 8051 with suitable examples of each.	L6	6
C) Draw structure of a TMOD register and explain function of each bit of it.	L5	6
 Q.4 Solve Any Two of the following.		 12
A) Write an 8051 C program to toggle all bits of P2 continuously every 500 ms. Use Timer 1, mode 1 to create the delay.	L4	6
B) Explain any six pins of RS232 DB-9 connector.	L1	6
C) Indicate which mode and which timer is selected for each of the following.	L3	6
i. MOV TMOD, #01H		
ii. MOVTMOD, #20H		
iii. MOV TMOD, #12H		
 Q. 5 Solve Any Two of the following.		 12
A) Write an 8051 C program to send letters 'A', 'B', and 'C' to the LCD using the busy flag method.	L1	6
B) Explain BSR and Input-Output modes of 8255 with neat diagrams.	L5	6
C) Explain internal RAM memory structure of an 8051 microcontroller.	L3	6

***** End *****